

This guide presents a set of practical MicroBlocks examples to introduce the main features of the LED Matrix Arabic MAX7219 library.

The library can:

- display Arabic Unicode text with automatic contextual letter shaping;
- control 1 to 8 chained MAX7219 8×8 LED matrices;
- support GENERIC and FC16 matrix modules;
- display variable-width Arabic and ASCII text;
- support Lam-Alif ligatures and Tatweel;
- display Arabic-Indic digits while preserving numerical order;
- align text to the Right, Center or Left;
- scroll Arabic, ASCII and Arabic-Indic text;
- combine Arabic, Arabic-Indic digits and non-Arabic text using Mixed groups;
- display Unicode icons using the native MicroBlocks font;
- apply dissolve, wipe, row wipe and curtain text effects;
- blink text with configurable ON/OFF times and repetitions.

Important

Run Arabic Matrix init before using the display blocks.

The examples in this guide use an ESP32 with the default pins:

CLK = 18, CS = 5, DIN = 23.

Other boards may require different pins.

Select the correct matrix type (GENERIC or FC16) and the number of chained devices for your hardware.

Hardware support

The library supports chains of 1 to 8 matrices, providing displays from 8×8 up to 64×8 pixels.

Hardware-tested configurations:

GENERIC: 1 and 4 matrices.

FC16: 1, 4 and 8 matrices.

Wiring

The examples use an ESP32 board with CLK = 18, CS = 5 and DIN = 23.

The wiring depends on the matrix module type and the number of chained matrices. Specific wiring diagrams are provided in the relevant examples.

For multi-module configurations, use an appropriate 5 V power supply for the LED matrices and ensure that the controller and matrix power supply share a common ground.

Arabic text

Arabic text is entered directly in Unicode. Contextual letter forms are handled automatically by the library. Lam-Alif and Tatweel are also supported.

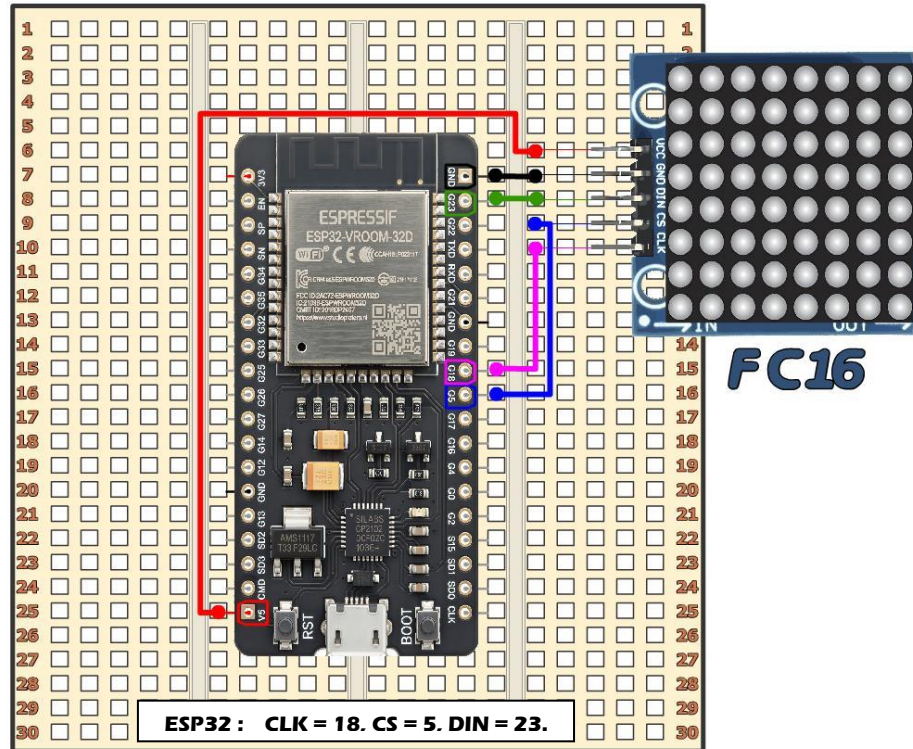
Example 1: Show Arabic Text.

Devices: 1 - Type: FC16 - Board: ESP32 - Pins: CLK 18, CS 5, DIN 23 - Rotation: 270°.

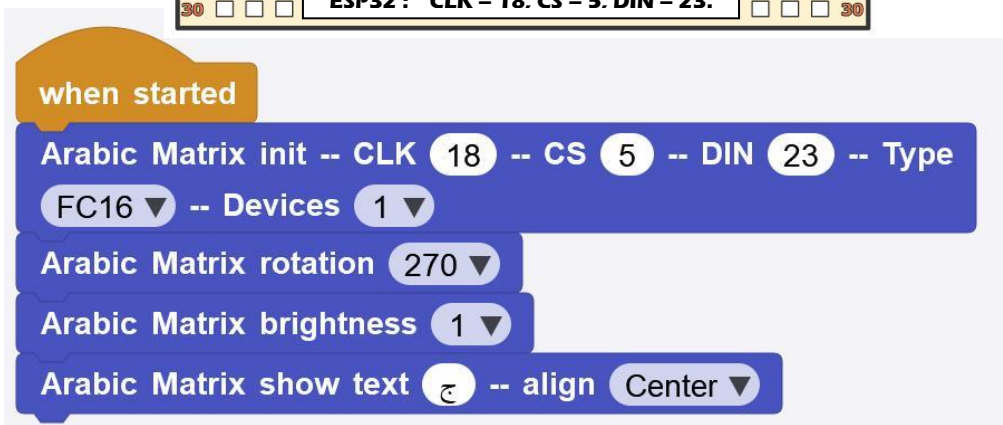
Objective

Learn the basic steps to initialize one FC16 MAX7219 8×8 matrix, set its rotation and brightness, and display a centered Arabic character.

Wiring



Program

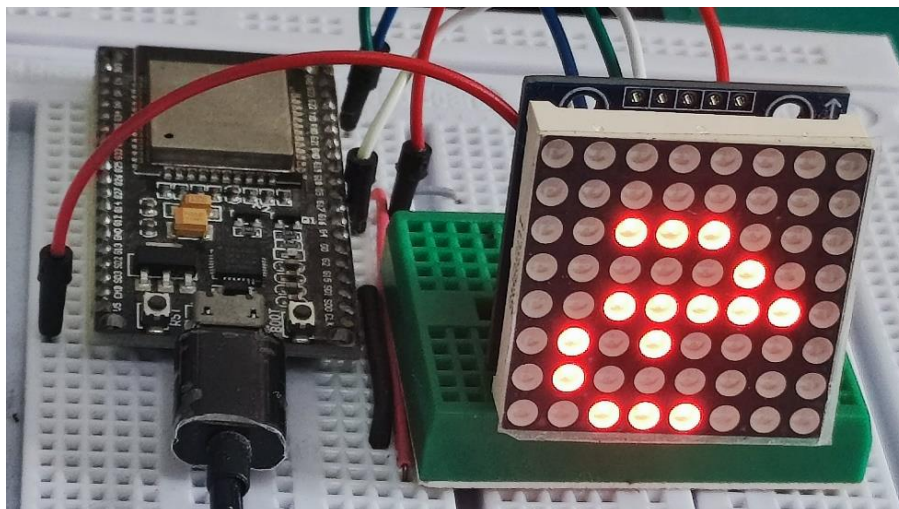


Note

Arabic text can be entered directly in Unicode. The library automatically selects the appropriate contextual form of supported Arabic letters.

In this example, the matrix is rotated by 270° to match its physical orientation.

Result



Example 2: Text Effects on Four GENERIC Matrices.

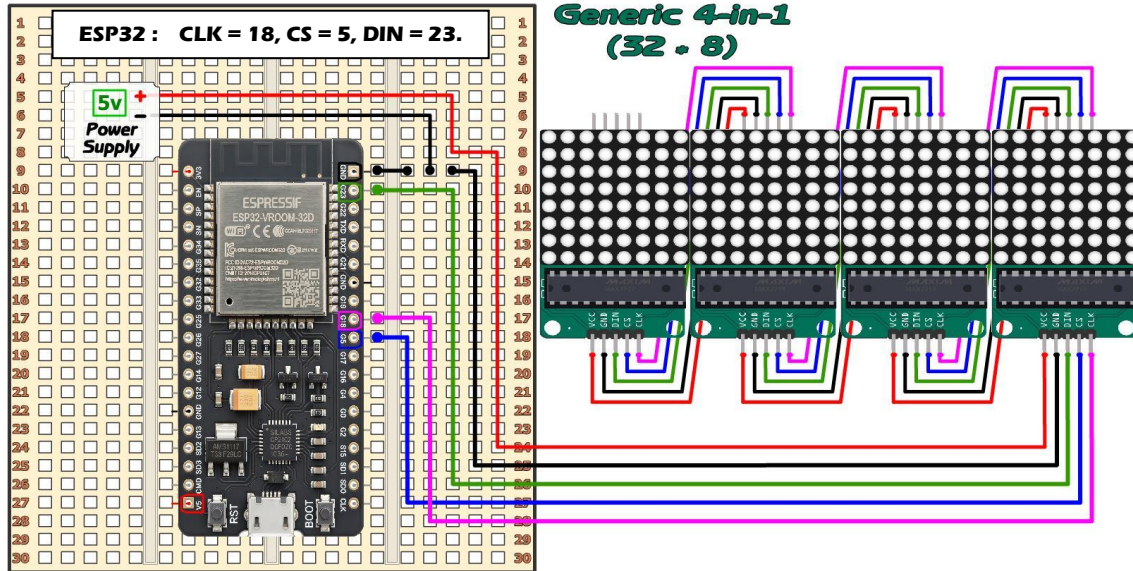
Devices: 4 - Type: GENERIC - Board: ESP32 - Pins: CLK 18, CS 5, DIN 23 - Rotation: 0°.

Objective

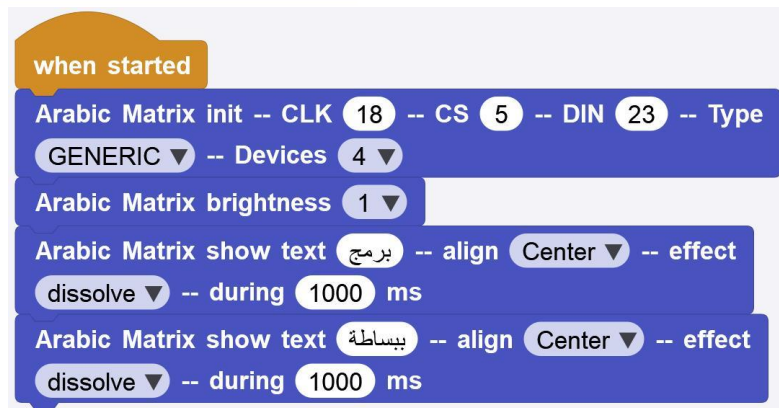
Display Arabic text across four chained GENERIC MAX7219 matrices and apply a text effect with Center alignment.

Wiring

This example uses four chained GENERIC matrices, providing a 32×8 pixel display. The LED matrices are powered from an external 5 V supply. The ESP32 and the matrix power supply must share a common ground.



Program

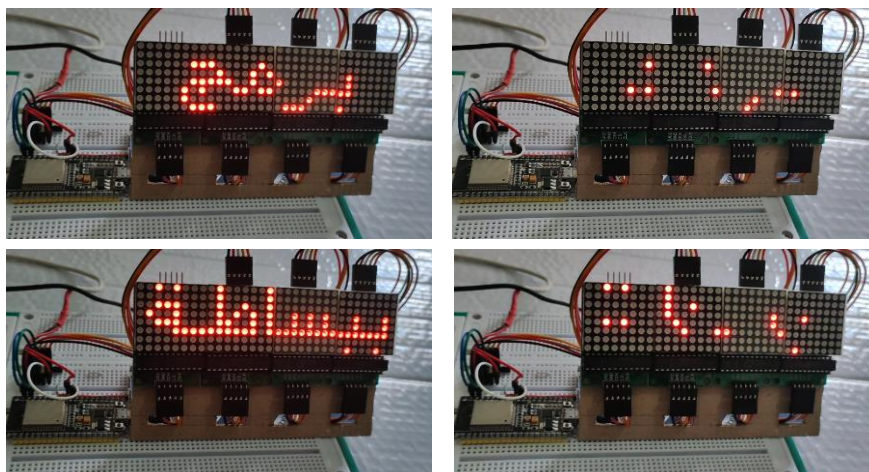


Note

The dissolve effect progressively reveals and removes the text while preserving Arabic shaping and the selected alignment.

This example also demonstrates a hardware-tested GENERIC ×4 configuration.

Result



Example 3: Show Mixed Text.

Devices: 8 - Type: FC16 - Board: ESP32 - Pins: CLK 18, CS 5, DIN 23 - Rotation: 0°.

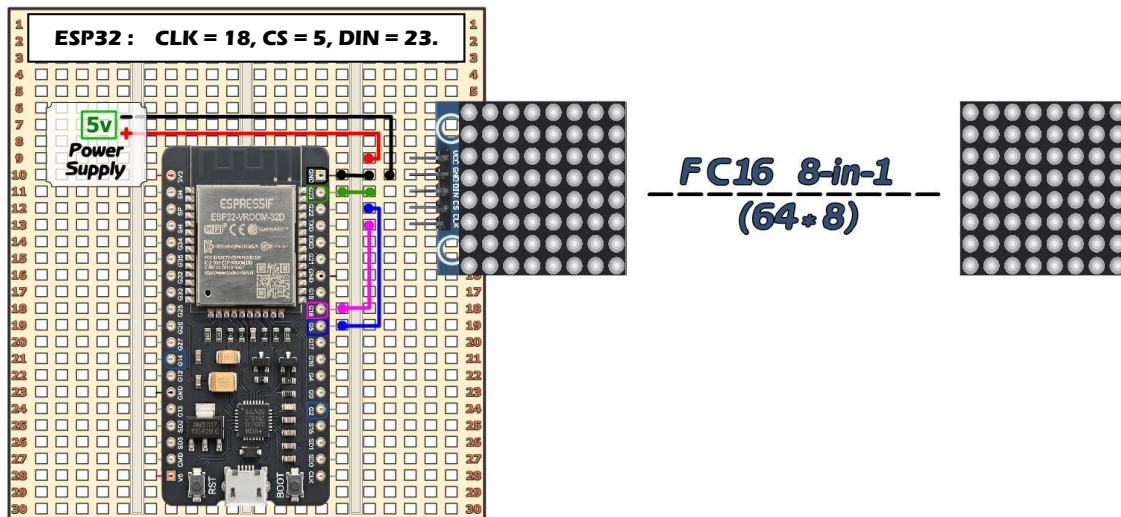
Objective

Combine Arabic and non-Arabic text on the same 64×8 display while preserving the appropriate composition direction for each group.

Wiring

Examples 3–7 use the same ESP32 and FC16 ×8 wiring shown below.

The eight matrices provide a 64×8 pixel display and use an external 5 V power supply with a common ground shared with the ESP32.



Program

when started

Arabic Matrix init -- CLK 18 -- CS 5 -- DIN 23 -- Type

FC16 -- Devices 8

Arabic Matrix brightness 1

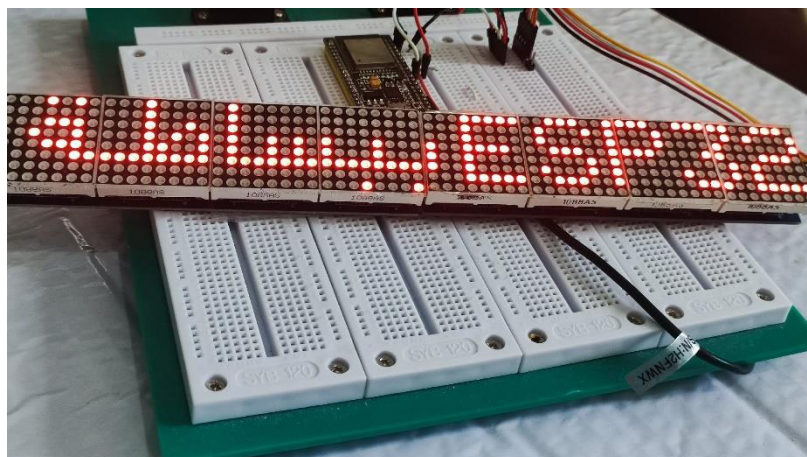
Arabic Matrix show mixed ESP32 ببساطة ▶

Note

Mixed text is organized into up to five independent groups. Group 1 is the rightmost group and Group 5 is the leftmost group.

Arabic groups use RTL composition, while non-Arabic groups use LTR composition. Empty groups are ignored, and a blank separation is automatically inserted between non-empty groups.

Result



Example 4: Blink Arabic Text.

Devices: 8 - Type: FC16 - Board: ESP32 - Pins: CLK 18, CS 5, DIN 23 - Rotation: 0°.

Objective

Display centered Arabic text with a configurable blinking sequence using independent ON and OFF durations and a selected number of repetitions.

Program

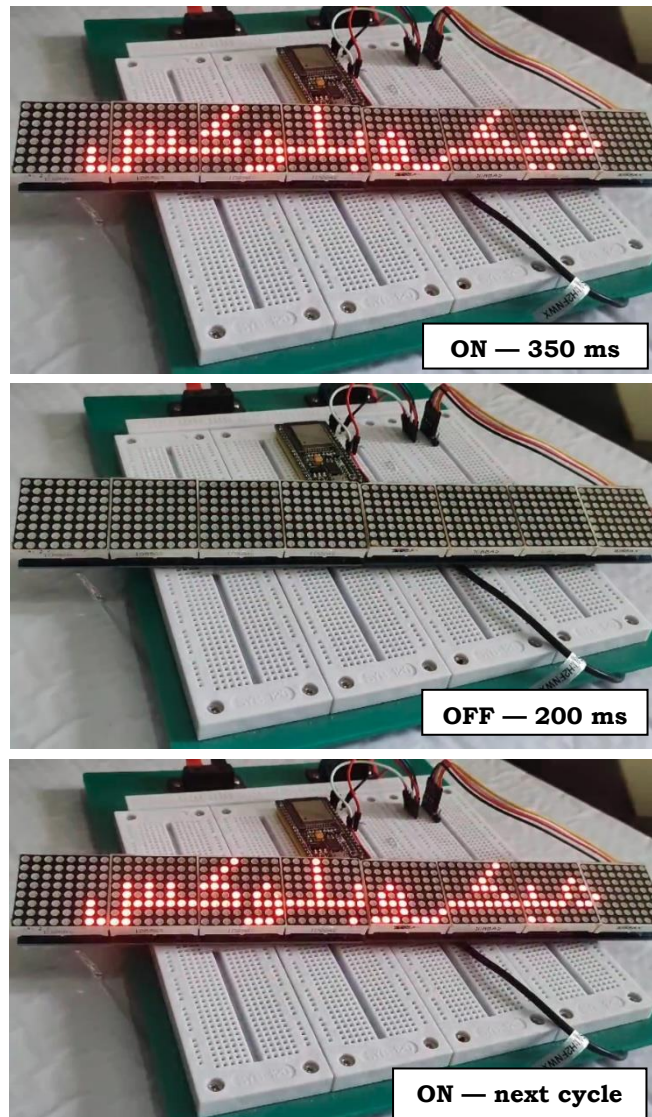


Note

The ON and OFF durations can be configured independently. The repeat value determines how many complete ON/OFF blink cycles are performed.

In this example, the text is displayed for 350 ms, cleared for 200 ms, and the cycle is repeated 5 times.

Result



Example 5: Arabic-Indic Digits.

Devices: 8 - Type: FC16 - Board: ESP32 - Pins: CLK 18, CS 5, DIN 23 - Rotation: 0°.

Objective

Convert ASCII digits to Arabic-Indic digits and display them in their correct numerical order using Mixed text.

Program



Note

The Arabic digits reporter converts ASCII digits 0–9 to Arabic-Indic digits ٠–٩ while preserving their numerical order.

In this example:

1234567890 → ١٢٣٤٥٦٧٨٩٠

The converted number is displayed as a Mixed group between two arrow icons. This makes the numerical direction easy to verify visually.

Arabic-Indic digits are handled independently from Arabic letter shaping and are displayed in LTR numerical order.

Result



Example 5: Scroll Arabic Text.

Devices: 8 - Type: FC16 - Board: ESP32 - Pins: CLK 18, CS 5, DIN 23 - Rotation: 0°.

Objective

Scroll a long Arabic message across an FC16 ×8 display using automatic scroll direction.

Program



Note

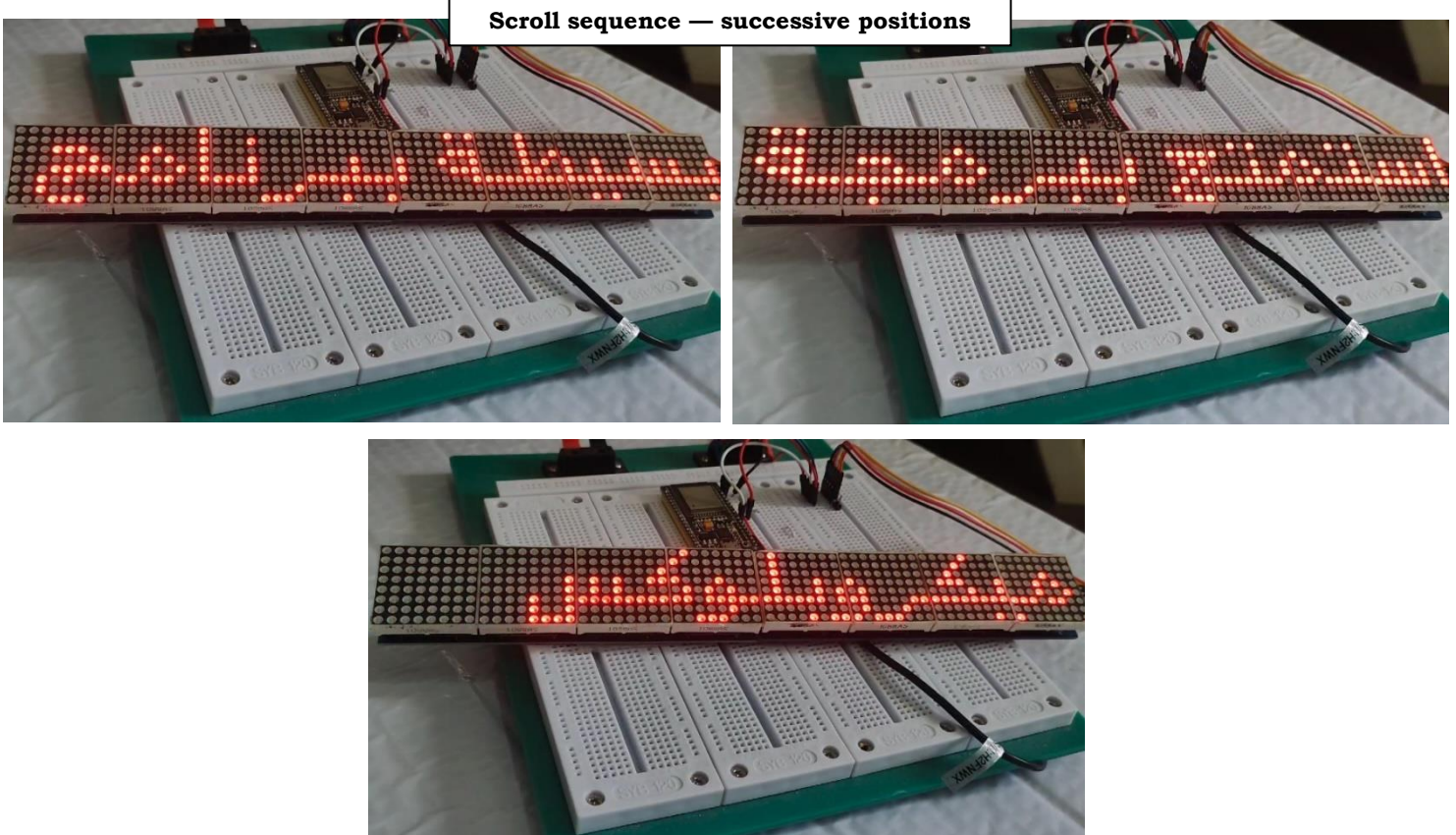
With direction set to Auto, the library automatically selects the normal scrolling direction according to the text type.

For Arabic text, Auto scrolls the composed text physically from left to right. The Arabic glyphs themselves are never mirrored.

The delay value controls the scrolling speed. Lower values produce faster movement. The library accepts values from 5 to 500 ms.

In this example, the message is scrolled three times with a delay of 25 ms.

Result



Example 7: Scroll Mixed Text.

Devices: 8 - Type: FC16 - Board: ESP32 - Pins: CLK 18, CS 5, DIN 23 - Rotation: 0°.

Objective

Scroll a mixed message containing Arabic text, ASCII text and a Unicode icon while preserving the composition direction of each group.

Program



Note

Mixed Scroll supports up to five independent groups. Group 1 is the rightmost group and Group 5 is the leftmost group.

Arabic groups preserve RTL composition, while non-Arabic groups preserve LTR composition. Empty groups are ignored, and a blank separation is automatically inserted between non-empty groups.

The Arabic icon reporter can be used as one of the Mixed groups. In this example, the Heart icon is rendered using the native MicroBlocks font.

The delay value controls the scrolling speed. Here, the mixed message scrolls with a delay of 30 ms.

The forever loop continuously repeats the complete Mixed Scroll sequence.

Result

